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(Autonomous)

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EXPT NO: 2	DATE: ________________

A PYTHON PROGRAM TO SOLVE THE TRANSPORTATION PROBLEM

AIM:
To solve the Transportation Problem for minimizing the total cost of distributing goods from multiple supply points to multiple demand points using Python.

PROCEDURE:
1. Install numpy and scipy using pip.
2. Define the cost matrix, supply list, and demand list.
3. Balance the problem by adding a dummy row or column if supply ≠ demand.
4. Formulate equality constraints for each supply and demand node.
5. Solve using linprog with method='highs' and display the optimal allocation and total cost.

PROBLEM 1: 3 Sources, 3 Destinations

CODE
```
import numpy as np
from scipy.optimize import linprog
def solve_transport(supply, demand, costs):
supply, demand = supply.copy(), demand.copy()
if sum(supply) > sum(demand):
demand.append(sum(supply) - sum(demand))
costs = np.hstack((costs, np.zeros((costs.shape[0], 1))))
elif sum(demand) > sum(supply):
supply.append(sum(demand) - sum(supply))
costs = np.vstack((costs, np.zeros((1, costs.shape[1]))))
m, n = len(supply), len(demand)
A = np.zeros((m + n, m * n))
for i in range(m):
A[i, i*n:(i+1)*n] = 1
for j in range(n):
A[m+j, j::n] = 1
b = np.array(supply + demand)
c = costs.flatten()
res = linprog(c, A_eq=A, b_eq=b, method='highs')
alloc = res.x.reshape(m, n)
return alloc, res.fun
supply = [30, 50, 20]
demand = [20, 40, 30]
costs = np.array([[2, 3, 1],[5, 4, 8],[5, 6, 8]])
alloc, total = solve_transport(supply, demand, costs)
print("Optimal Allocation:")
```

```
print(alloc)
print("Total Cost:", total)
```

OUTPUT

```
Optimal Allocation:
[[ 0. 0. 30.]
[20. 30. 0.]
[ 0. 10. 0.]]
Total Cost: 200.0
```

PROBLEM 2: 2 Sources, 4 Destinations

CODE

```
supply = [50, 60]
demand = [30, 20, 25, 35]
costs = np.array([[3, 2, 7, 6],[2, 5, 1, 3]])
alloc, total = solve_transport(supply, demand, costs)
print("Optimal Allocation:")
print(alloc)
print("Total Cost:", total)
```

OUTPUT

```
Optimal Allocation:
[[ 0. 20. 0. 30.]
[30. 0. 25. 5.]]
Total Cost: 200.0
```

PROBLEM 3: 4 Sources, 3 Destinations (unbalanced)

CODE

```
supply = [120, 80, 80, 60]
demand = [150, 80, 60]
costs = np.array([[4, 8, 1],[7, 2, 3],[3, 9, 2],[9, 3, 6]])
alloc, total = solve_transport(supply, demand, costs)
print("Optimal Allocation:")
print(alloc)
print("Total Cost:", total)
```

OUTPUT

```
Optimal Allocation:
[[ 0. 0. 60. 60.]
[ 0. 80. 0. 0.]
[80. 0. 0. 0.]
[60. 0. 0. 0.]]
Total Cost: 550.0
```

RESULT:

Thus, the Transportation Problem for minimizing the cost of distributing goods using Python has been implemented and executed successfully.
